Morgan M. Hardjadinata

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PROFESSIONAL SUMMARY

Data Science and AI Engineering student at UW-Madison with hands-on internship experience at one of the largest automotive semiconductor manufacturers. Specialized in developing and deploying multimodal, agentic AI systems using LLMs, RAG, and DevOps principles. Proven ability to deliver production-level AI solutions that automate complex diagnostics and optimize supply chain analytics, seeking a challenging full-time AI/ML Engineer role.

EDUCATION

University of Wisconsin-Madison **Madison, WI**

Bachelor of Science, Computer and Data Science | Dec 2025 GPA: 3.64/4.0

Relevant Coursework: Artificial Intelligence, Big Data Systems, Database Management, Operating Systems, Algorithms & Data Structures, Linear Algebra, Data Science Modeling.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, R, HTML, CSS, JavaScript.
- **AI/ML Frameworks:** PyTorch, LangChain, LangGraph, Autogen, Hugging Face, LlamaIndex
- **Cloud/DevOps:** AWS, GCP (Vertex AI), Kubernetes, Docker, Spark, Hadoop
- **Databases:** ChromaDB, SQL, Silicon.da

EXPERIENCE

Infineon Technologies **Tokyo, Japan & San Jose, CA**

AI & Embedded Software Engineer *June 2025 – Sep 2025*

- Led the implementation of a **Multimodal AI** agent automating SRAM/SCAN diagnostics for **10+** Traveo & PSoC microcontrollers, achieving an estimated **40% reduction** in manual analysis time.
- Engineered a critical **data migration pipeline** for the supply chain department, transitioning **over 5 TB** of legacy data to the AI-ready Silicon.da cloud platform to enable new Gen AI analytics.
- Developed and deployed LLM agentic frameworks (**LangChain, CrewAI**) to process and analyze mass production data, identifying key manufacturing trends to support a **5% optimization** in supply chain efficiency.

Infineon Technologies **Singapore**

Software Engineer *May 2024 – Sep 2024*

- Developed and deployed an AI-powered automation solution for the Electrical Failure Analysis (EFA) lab, increasing overall lab productivity by **30%** for the Tricore product line.
- Engineered a **Hybrid Search Retrieval-Augmented Generation (RAG)** system on Google's Vertex AI to process over **500 GB** of unstructured technical documents, improving the accuracy of AI-generated diagnostic reports by **50%.**
- Built a unified AI interface that automated the concurrent control of **six** distinct electrical test systems, reducing manual operator intervention by **40%** and minimizing human error.

PROJECT EXPERIENCE

Bellevue College Programming Club, Bellevue, WA Jun 2023 – Dec 2023

Developed a full-stack web application using Python (Flask) and JavaScript (React) to help **200+** students plan their course schedules, pulling real-time data from the college's public API.

EXTRACURRICULAR

Badger Solar Racing – *Firmware CAR 2 and Website Operator* May 2025 – Present

Designed and engineered an interactive steering wheel dashboard UI to provide real-time data visualization (e.g., speedometer, system diagnostics) and control over the vehicle's hydraulics, directly contributing to the optimization of in-race power consumption.

Badger Drone Club – *Computer Vision* July 2024 – Sep 2025

Developing computer vision systems with OpenCV and Python to implement autonomous object tracking and a hand gesture recognition module for intuitive, real-time drone command and control of DJI Tello Drones.